

packages/graph-core/src/fingerprint.ts

Kind: added · Baseline: a9c34bfb03fe · Target: NovaDiff · Generated: 2026-05-20T05:24:43.590Z

Overview

A new module `packages/graph-core/src/fingerprint.ts` implements a fingerprinting pipeline for JavaScript/TypeScript files. It hashes file contents, extracts structural signatures (functions, classes, imports, exports), and classifies changes into NONE, COSMETIC, or STRUCTURAL.

Key changes

- **Imports** (R1-R5): added `node:crypto`, `node:fs`, `node:path`, and type imports from `./types.js` and `./plugins/registry.js`.
- **Type definitions** (R9-R48): introduced `FunctionFingerprint`, `ClassFingerprint`, `ImportFingerprint`, `FileFingerprint`, `FingerprintStore`, and `ChangeLevel`.
- **Helpers** (R70-R121): `contentHash` computes SHA-256; `extractFileFingerprint` builds a `FileFingerprint` from a `StructuralAnalysis`.
- **Comparison** (R131-R246): `compareFingerprints` checks content hash, structural signatures, and flags significant size changes. Missing structural analysis defaults to STRUCTURAL.
- **Store builder** (R253-R291): `buildFingerprintStore` reads files, runs `registry.analyzeFile`, and falls back to hash-only fingerprints when analysis is unavailable.
- **Change analyzer** (R297-R385): `analyzeChanges` orchestrates file existence checks, fingerprint extraction, and categorizes changes into new, deleted, unchanged, cosmetic, or structural.

Impact

- **Correctness:** Structural comparison can surface API changes that content hashing alone would miss.
- **Maintainability:** Centralized fingerprint logic replaces scattered implementations.
- **Compatibility:** The module is additive; no existing APIs are altered.
- **Performance:** Reading and hashing all files adds runtime cost; impact on CI or hot-reload scenarios is unknown from the diff.

Risks & follow-ups

- **False positives:** `JSON.stringify` on arrays may mis-order elements; ensure `methods / properties` are sorted before comparison.
- **Large file handling:** `content.split("\n")` could be memory intensive; benchmark on large files.
- **Registry contract:** `registry.analyzeFile` must return `StructuralAnalysis` or `null`; otherwise fingerprints become hash-only.

- **Test coverage:** Add unit tests for `compareFingerprints` edge cases (e.g., no structural analysis, significant size change).